

# SCION: The Structural Intelligence Engine

Transitioning Teradata DNA from Rented Capability to Owned Intelligence.

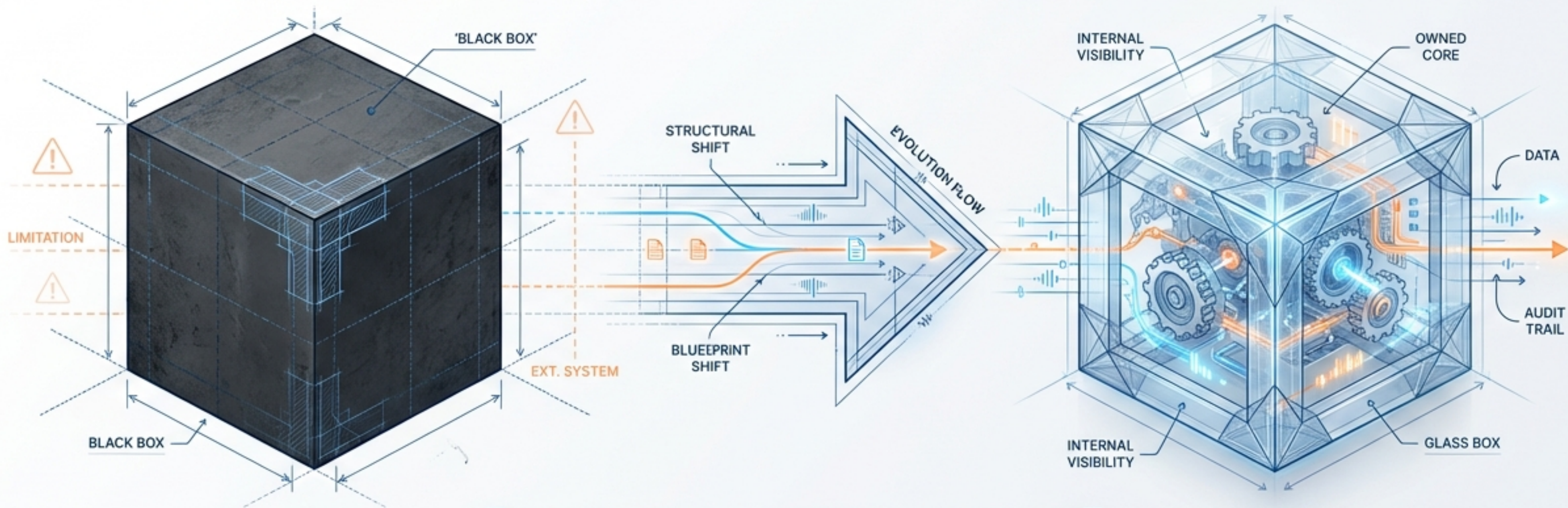




# The Transition: Owning Our Structural Intelligence

**FROM: Rented Capability**

**TO: Owned Intelligence**



• External Dependency



• Opaque Logic



• High Operational Expense



• Internal Asset



• Transparent & Auditable

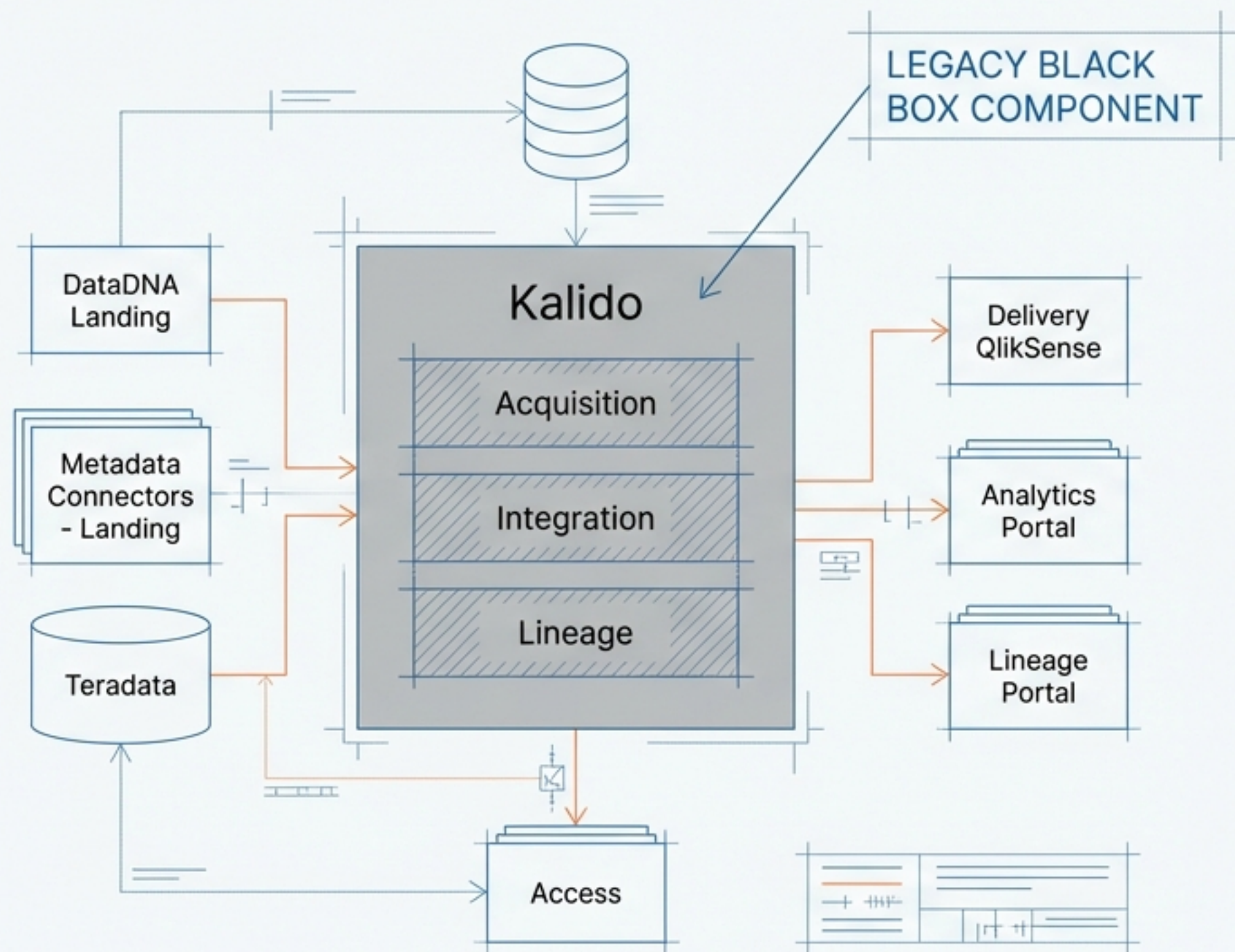


• Strategic Intellectual Property

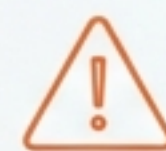




# The Current Landscape: Limitations of the Black Box



## The Problem with the Status Quo



- **High Recurring OpEx:** Significant licensing costs for a capability we simply "rent".



- **Black-Box Logic:** Lineage and parsing rules are hidden; we cannot explain to clients *how* results are derived.



- **Vendor Dependency:** Agility is throttled by a third-party roadmap, not our own.



- **Limited Integration:** Closed architecture prevents deep integration with modern AI reasoning (TAISA).

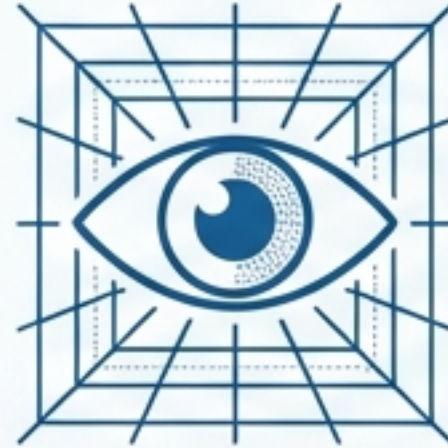


# The Objective: Why We Build Instead of Buy



## Cost Control

Replace recurring license fees with a one-time development investment.



## Transparency

Move to a "Glass Box" model where 100% of the logic is explainable and auditable.



## Strategic IP

Transform an operational expense into a permanent, owned asset for Teradata.



## Agility

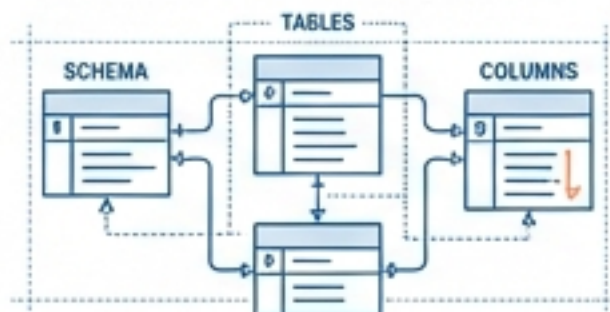
A decoupled architecture that evolves at our speed, enabling rapid adaptation.



# Defining SCION

# S

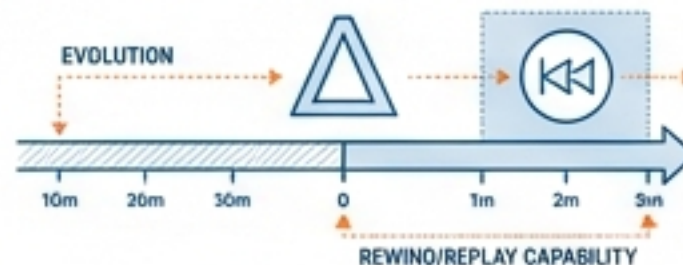
**S — Structural**



Focuses on the skeleton of the data (schema, tables, columns), not the business content.

# C

**C — Change**



Tracks exactly how platforms evolve over time (rewind/replay capability).

# I

**I — Intelligence**



Understands what changed and implies the “why” via automated reasoning.

# O

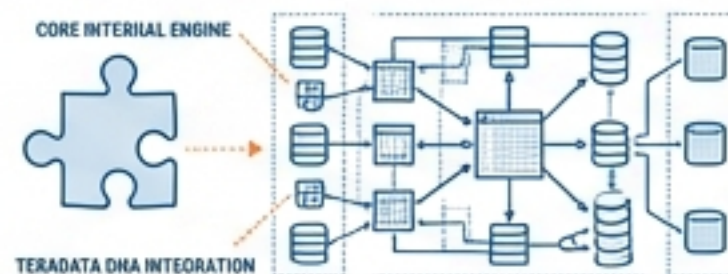
**O — Observability**



Makes hidden structural shifts visible, traceable, and auditable.

# N

**N — Node**

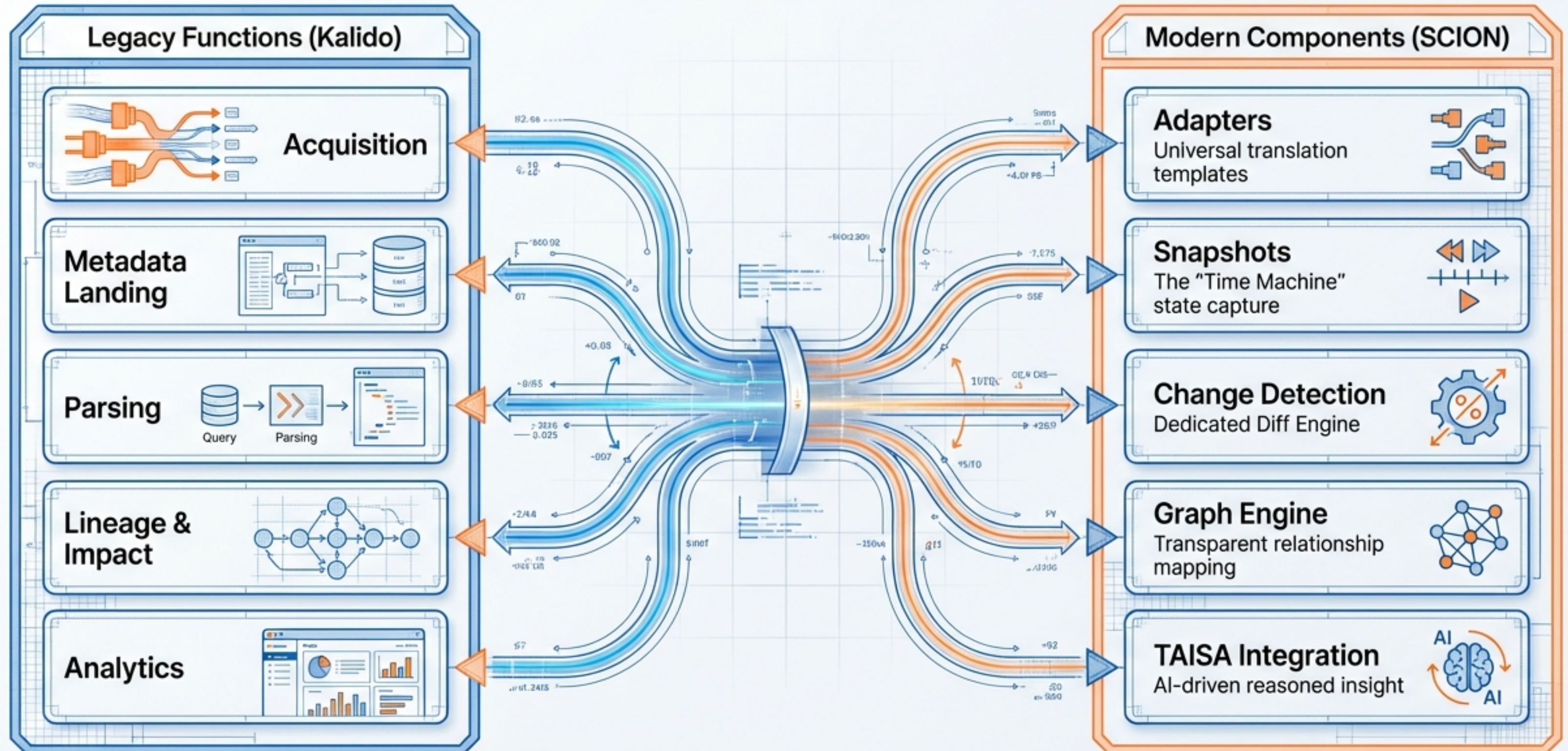


Acts as a core internal engine \*within\* Teradata DNA, not a standalone product.



# Functional Mapping: Kalido vs. SCION

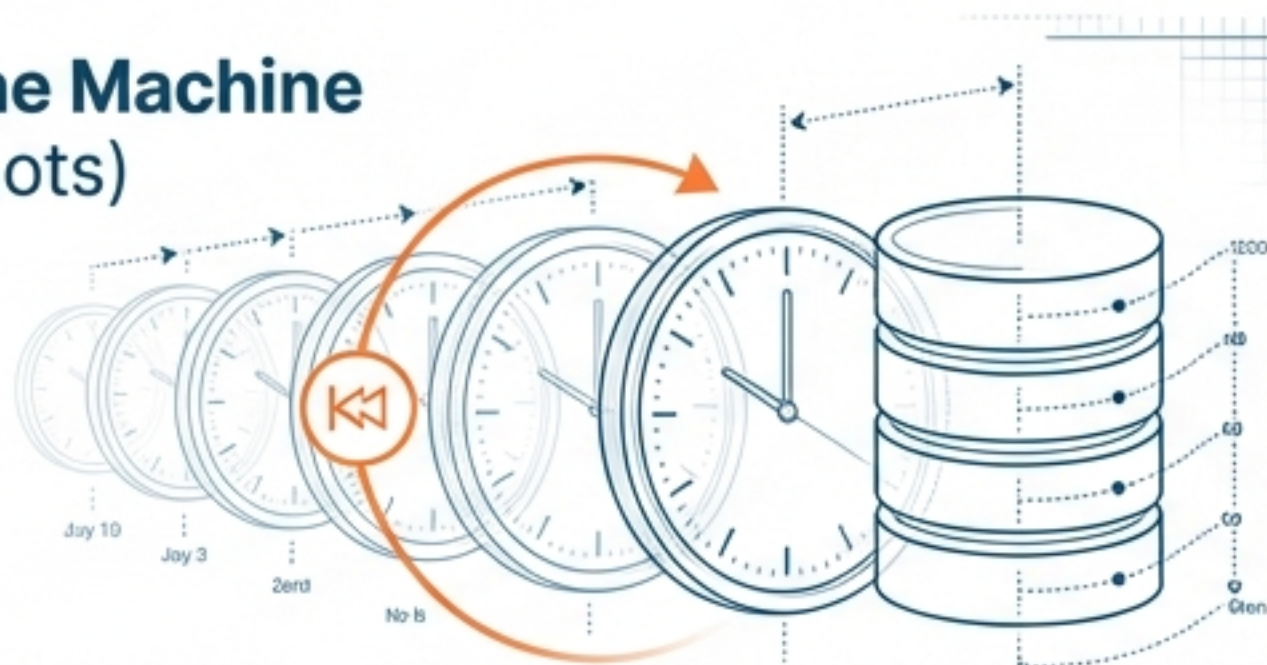
Modernizing responsibilities with a modular architecture





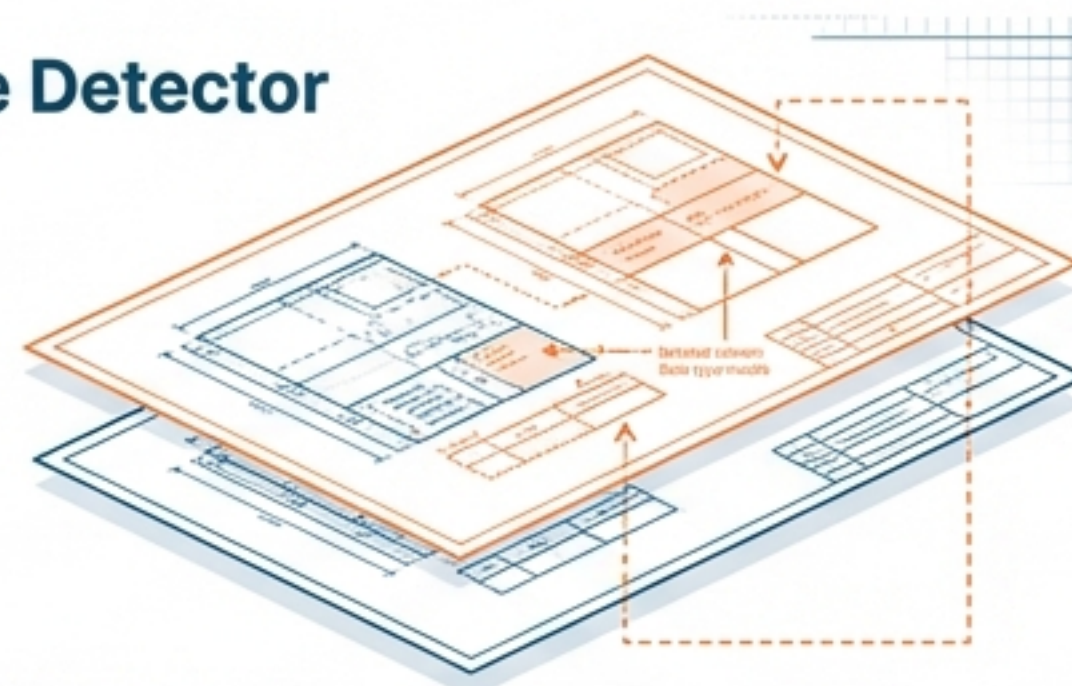
# Under the Hood: Key Components

## The Time Machine (Snapshots)



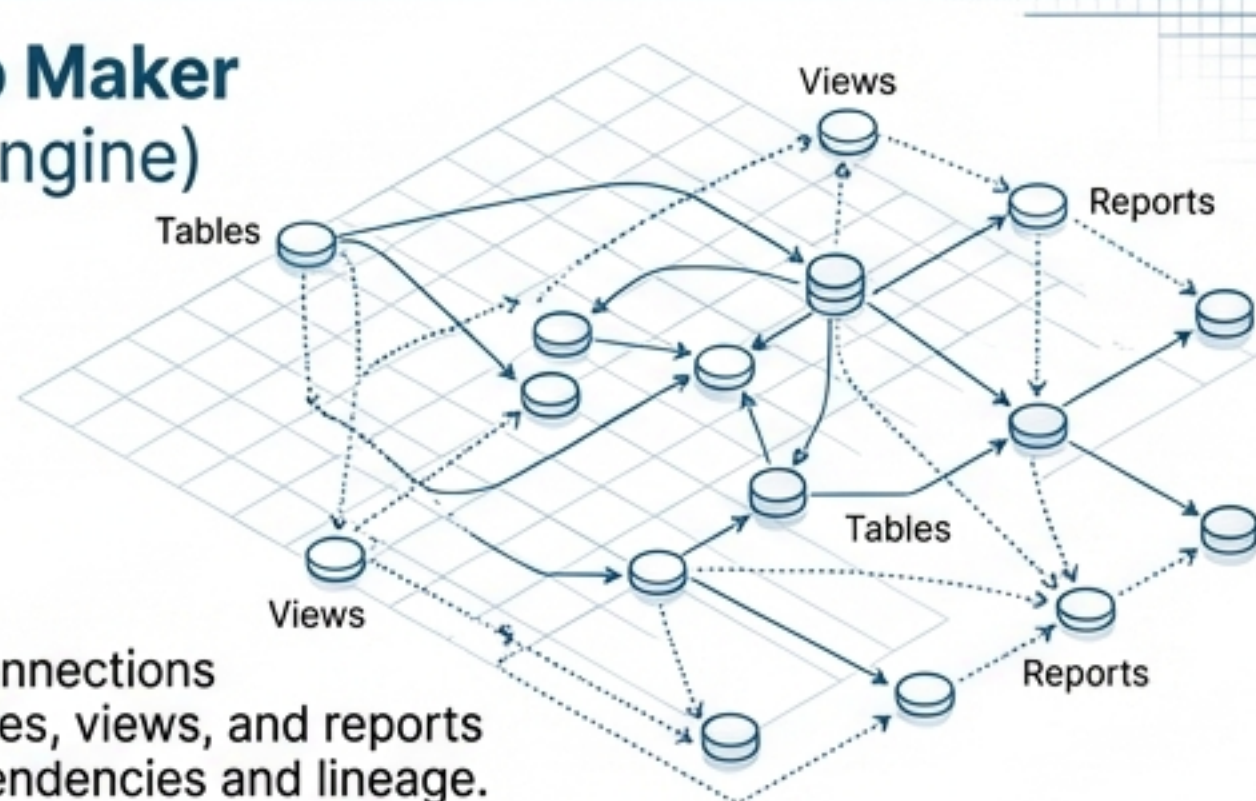
Captures the exact state of the data platform at any moment.  
Allows us to "rewind" structure to see yesterday's reality.

## The Difference Detector (Diff Engine)



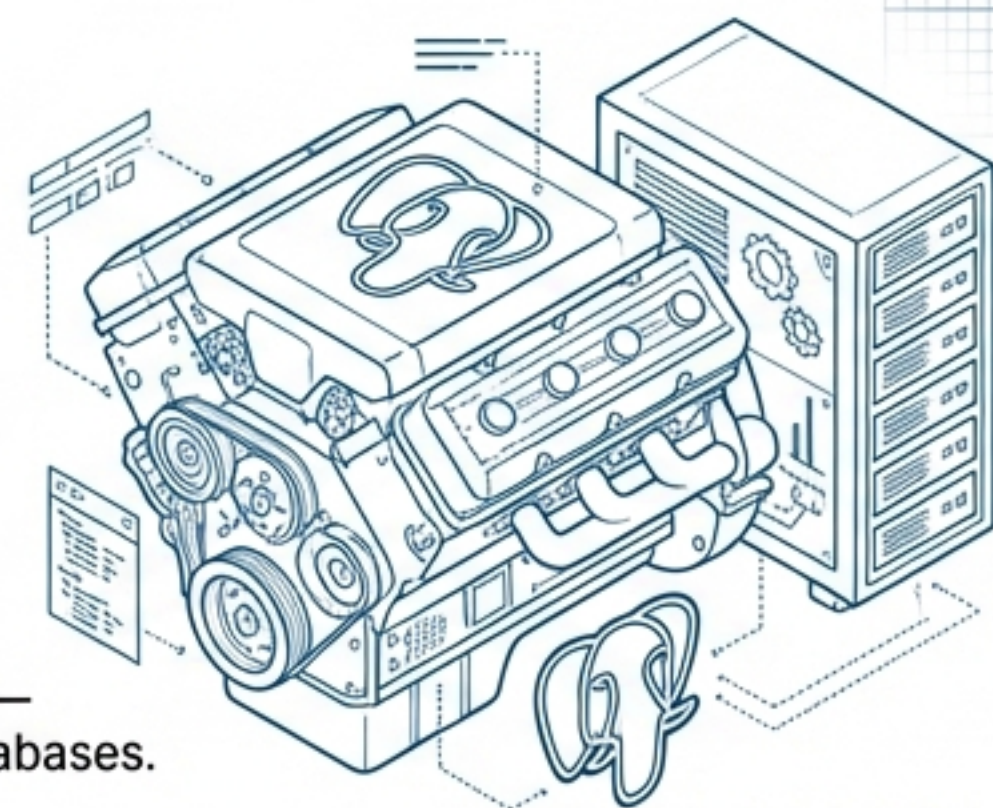
Automatically spots what changed between two points in time  
(e.g., a column drop or type change).

## The Map Maker (Graph Engine)



Draws the connections  
between tables, views, and reports  
to show dependencies and lineage.

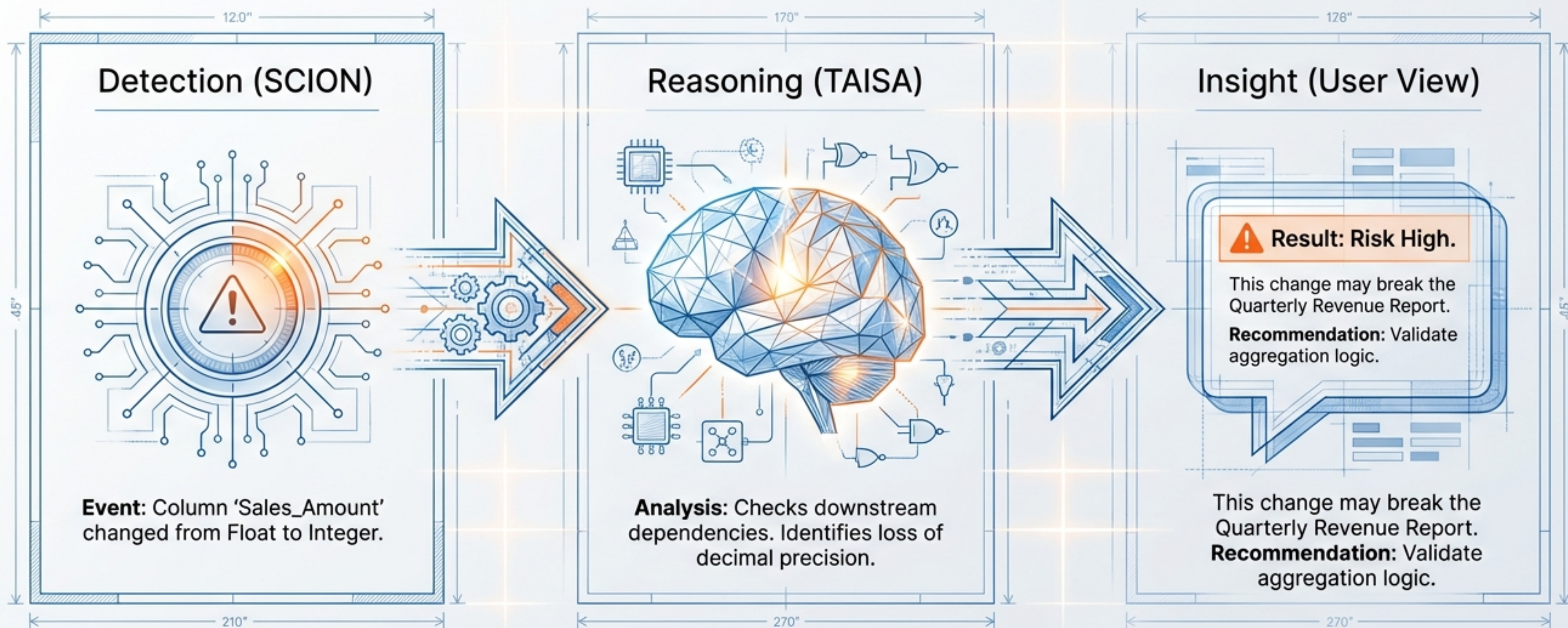
## The Core (PostgreSQL)



Built on standard, robust  
open-source technology—  
no exotic proprietary databases.



# SCION + TAISA: From Detection to Reasoning



The advisor that explains *why* a change is risky, not just that it happened.



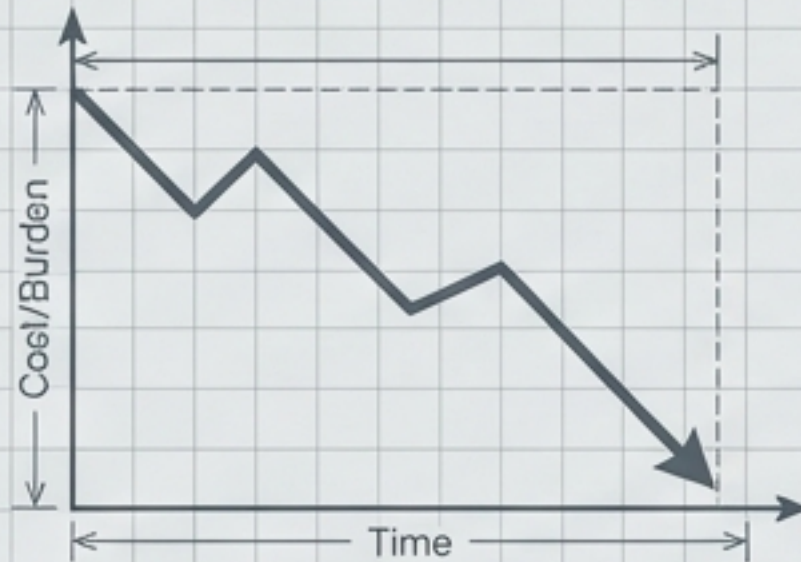
# Comparative Analysis

| Dimension    | Legacy (Kalido)         | SCION (Internal)                                                                                                        |
|--------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Transparency | Black Box (Opaque)      | Glass Box (Fully Explainable)        |
| Flexibility  | Vendor-Locked Roadmap   | Internally Owned & Adaptable         |
| Cost Model   | High Recurring Licenses | One-time Investment                |
| Intelligence | Static Rules            | Reasoned Insight (AI/TAISA)        |
| Ownership    | Rented Capability       | Proprietary Intellectual Property  |

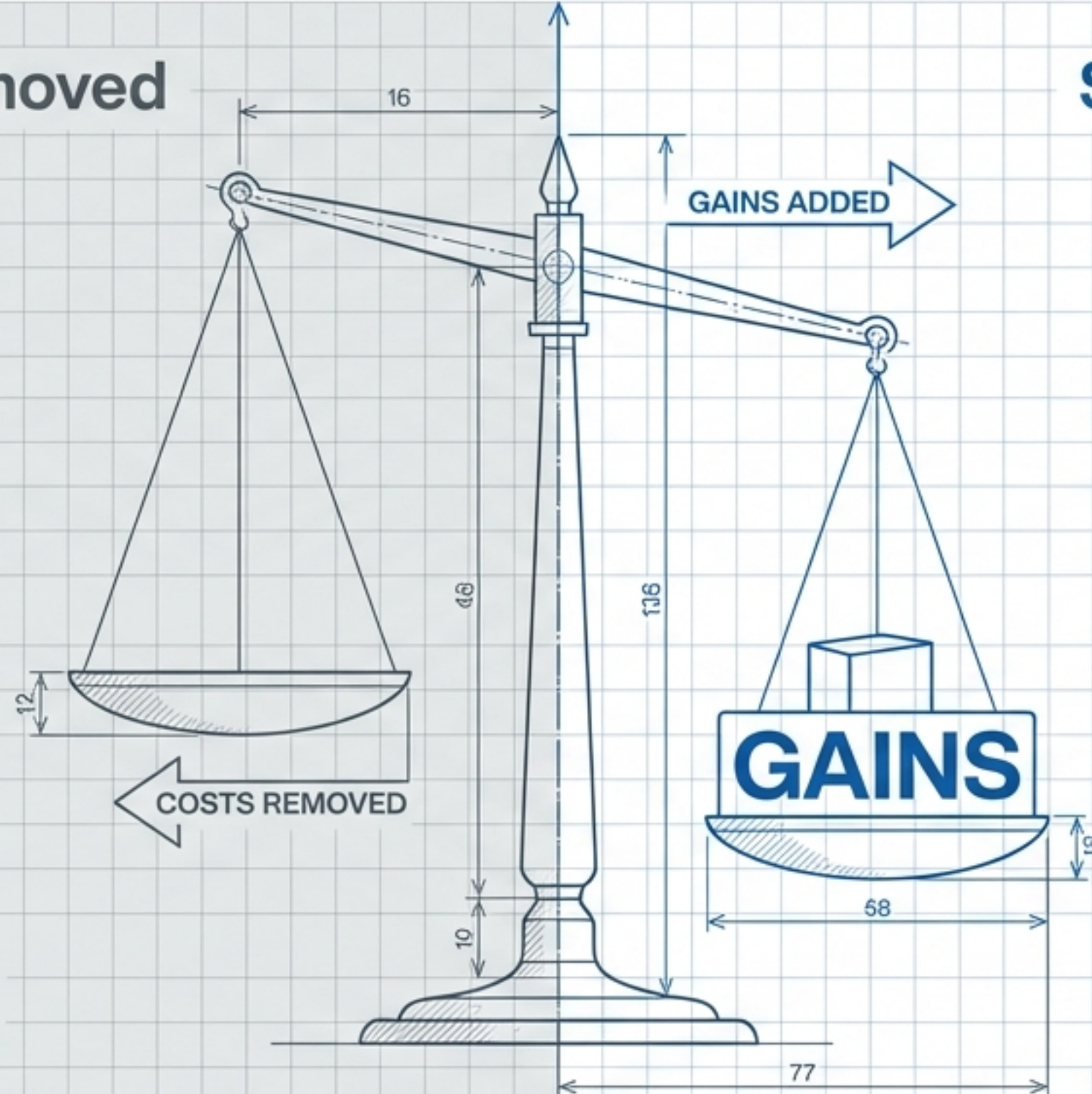


# Cost vs. Gain Analysis

## Costs & Burdens Removed



- Elimination of Kalido licensing fees.
- Reduced operational overhead for debugging 'black box' issues.
- Removal of dependency on third-party roadmaps.



## Strategic Gains Added



- Creation of permanent Intellectual Property. 
- Faster time-to-market for new connectors (Snowflake, BigQuery). 
- 'White box' auditability for client trust. 



# Defining the Scope: What SCION is NOT

## Not a BI Tool

It does not replace dashboards like Qlik or Tableau.

## Not an ETL Tool

It tracks structure; it does not move or transform business data.

## Not a Business Glossary

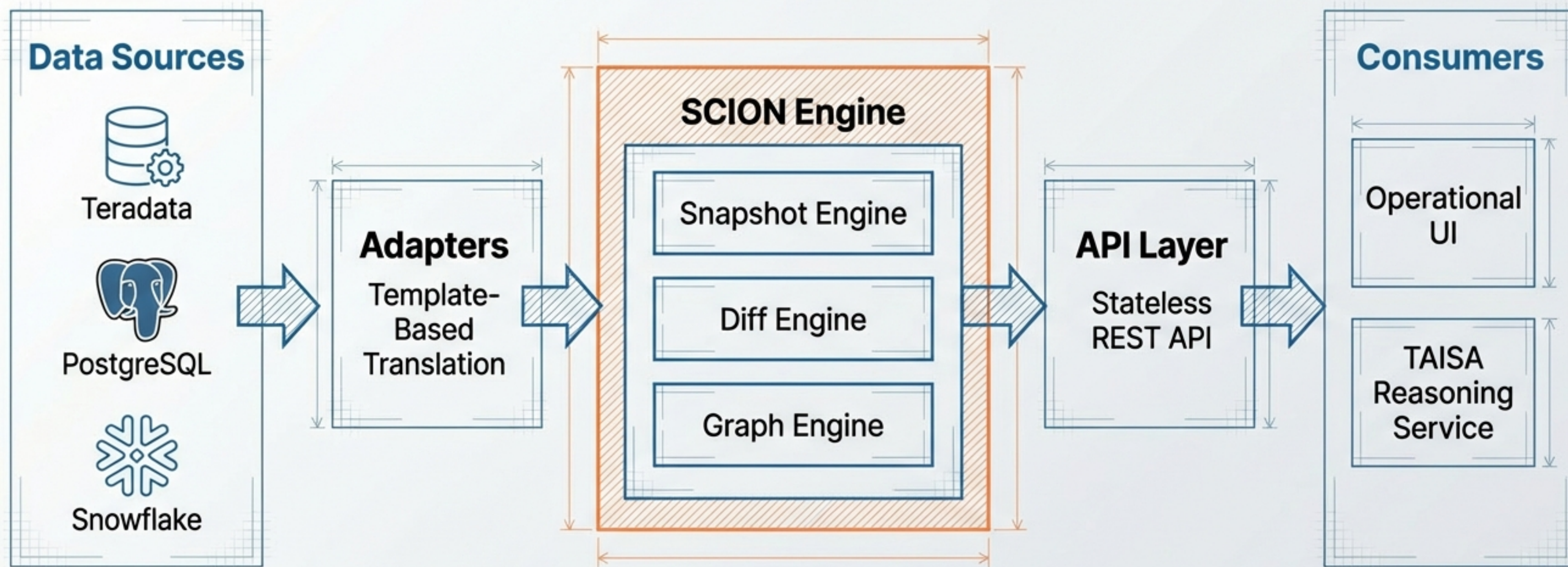
It handles technical metadata, not business definitions.

## Not a Standalone Product

It is a specialized component  
*\*inside\** Teradata DNA.



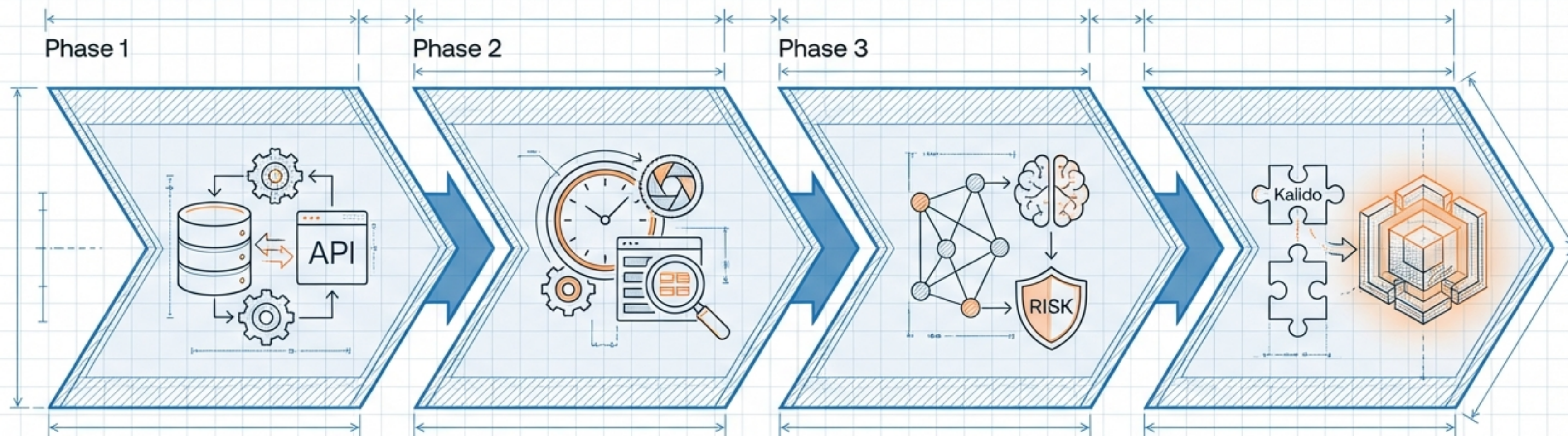
# Technical Architecture Overview



Deployed via Docker • API-First Design • Stateless Intelligence



# High-Level Roadmap



## Phase 1 Foundation

Setup of internal PostgreSQL engine, API, and basic Adapters.

## Phase 2 Intelligence

Enabling the “Time Machine” (Snapshots) and “Difference Detector” (Change Detection).

## Reasoning

Integrating the Graph Engine (Lineage) and TAISA for risk analysis.

## Integration

Full replacement of the Kalido component within the Teradata DNA ecosystem.



# The Future of Teradata DNA

“SCION is not just a replacement for Kalido. It is a strategic shift from rented capability to owned intelligence.”

